

Towards StatLib: A Tour of Formal Mathematics for Statisticians

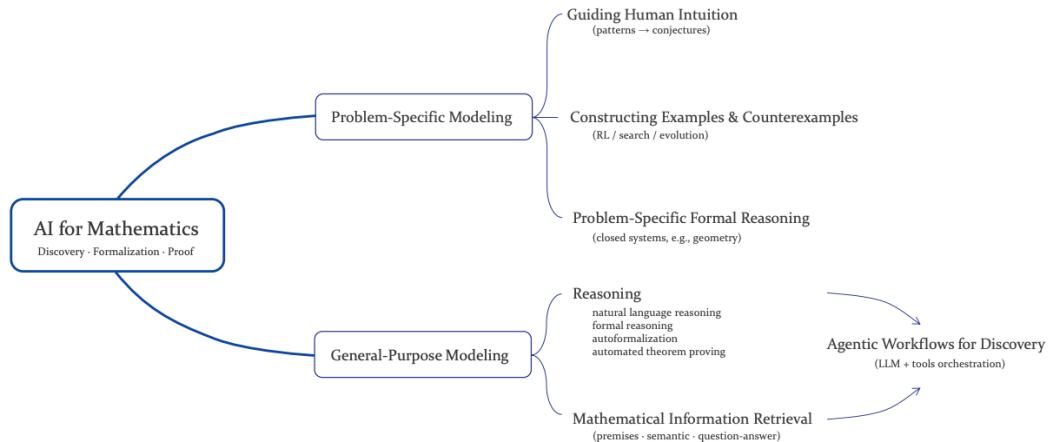
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I. AI for Math

Landscape of AI4Math research directions [1]



Machine-Assisted Proof [2]

Mathematicians have used **machines** and **computers** to assist them in their research for centuries.

One of the earliest uses of (human) computers in mathematics was to build tables, such as **logarithm tables** and **trigonometric tables**.

Modern tables are still important today, particularly in **experimental mathematics**.

- In the late 18th century, Adrien-Marie Legendre and Carl Friedrich Gauss used tables of prime numbers to conjecture the **prime number theorem**.
- In the 1960s, Bryan Birch and Peter Swinnerton-Dyer used tables of elliptic curve data to formulate the **Birch and Swinnerton-Dyer conjecture**.

— after Terence Tao, AMS Notices 2025; presentation after Leonardo de Moura.

In 2026, **Boris Alexeev & Dustin G. Mixon** resolve a **\$1000 Erdős prize problem**: **ChatGPT** generates the Lean proof, and **Lean** verifies it. “We used ChatGPT to **vibe code** a Lean proof.”

Without the verifier, “an AI said so” is not mathematics — **the proof is machine-checked**.

What is Lean?

Lean is an open-source *programming language* and *proof assistant* that is transforming how we approach mathematics, software verification, and AI.

- **Public debut:** CADE 2015 — Berlin.
- **Self-hosted.** Lean and its tooling are *implemented in Lean*; the system is highly extensible.
- **Small trusted kernel.** Proofs can be **exported and independently checked**.

“Theorem Proving Is a Game.”

— Kevin Buzzard & Leonardo de Moura

Ready for fun? Let's code!

The Liquid Tensor Experiment

In 2020, **Peter Scholze** posed a formalization challenge.

“I spent much of 2019 obsessed with the proof of this theorem, almost getting crazy over it. I still have some small lingering doubts.”

— Peter Scholze

Johan Commelin led a team that verified the proof, with only minor corrections. **They verified and simplified it without fully understanding the proof.**

“The Lean Proof Assistant was really that: an assistant in navigating through the thick jungle that this proof is.”

— Peter Scholze

Equational Theories Project (Terence Tao) · **Carleson’s Theorem** (Floris van Doorn) · **Sphere packing in dim 8 & 24** (Maryna Viazovska; AI-formalized by Math, Inc.’s Gauss agent) · **Fermat’s Last Theorem** (Kevin Buzzard) · **PFR conjecture** (Timothy Gowers, Ben Green, Freddie Manners, Terence Tao) · **Prime Number Theorem** (Terence Tao & Alex Kontorovich).

II. What made it work

Mathlib — Digital Foundation of Mathematics

Mathlib: an open-source library of formal mathematics written using the **Lean theorem prover**.

- Open-source, collaborative.
- 770+ contributors, ~ 2.1M lines of code.
- ~ 275,000 theorems, ~ 131,000 definitions.

"I'm investing time now so that somebody in the future can have that amazing experience."

— Heather Macbeth

What's in 2.1M lines of formal math?

How many of them are **mathematical statistics**?

<https://leanprover-community.github.io/mathlib-overview.html>

Towards Statlib

Hopefully we have convinced you we need a **statistical library**.

Let's take one example: **how would you consider the definition of risk?**

Statistical decision theory — the four stages

A fundamental premise of statistical inference: decision theory provides **rigorous mathematical language** for optimality and accuracy.

- 1 An **experiment** to understand an **object of interest**.
- 2 Given observations, a **class of actions / decisions** to quantify the understanding.
- 3 **Calibrating** the precision of the understanding.
- 4 Exploring the **“best” decision rule**.

Why Markov kernels?

Every transition — parameter $\rightarrow$ randomness $\rightarrow$ observation $\rightarrow$ action $\rightarrow$ loss $\rightarrow$ risk — is a **Markov kernel**, a pushforward, or an integral against one. The natural framework is kernel-first.

Markov kernel — the definition

Given two measurable spaces $(Z_1, \mathcal{M}_1)$ and $(Z_2, \mathcal{M}_2)$, a **Markov kernel**

$$K : (Z_1, \mathcal{M}_1) \rightarrow (Z_2, \mathcal{M}_2)$$

is a map $K : Z_1 \times \mathcal{M}_2 \rightarrow [0, 1]$ such that

- for all $z_1 \in Z_1$, $K(z_1, \cdot)$ is a **probability measure** on $(Z_2, \mathcal{M}_2)$;
- for all $A \in \mathcal{M}_2$, $K(\cdot, A)$ is a **measurable function** on $(Z_1, \mathcal{M}_1)$.

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example (f : X → Measure Y) (hf : Measurable f) : Kernel X Y := ⟨f, hf⟩
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Components

Sequence indexed by $n \in \mathbb{N}$.

- Measurable spaces $(\Theta_n, \mathcal{A}_n)$ (parameter space), $(\Omega_n, \mathcal{G}_n)$ (space for experimental randomness).
- **Markov kernel** $K_n : (\Theta_n, \mathcal{A}_n) \rightarrow (\Omega_n, \mathcal{G}_n)$ to drive experimental randomness given its trying to measure Θ_n .
- Measurable space $(S_n, \mathcal{F}_n)$ (functional / object of interest space) and a measurable map $\Psi_n : \mathcal{K}_n \rightarrow S_n$, where $\mathcal{K}_n$ is a subset of Markov kernels equipped with the σ -algebra that makes all point evaluations $K_n(\cdot, A)$ measurable.
- Measurable space $(\mathcal{X}_n, \mathcal{B}_n)$ (observation space) and $\mathcal{X}_n$ -valued observation $X_n : \Omega_n \rightarrow \mathcal{X}_n$, a measurable map.
- Measurable space $(\mathcal{Y}_n, \mathcal{C}_n)$ (decision space) and **Markov kernel** $T_n : (\mathcal{X}_n, \mathcal{B}_n) \rightarrow (\mathcal{Y}_n, \mathcal{C}_n)$ (randomized action given observation).
- $\ell_n : \mathcal{Y}_n \times S_n \rightarrow [0, +\infty]$ (loss function) jointly measurable w.r.t. the product σ -algebra, and $\ell_n(x_n, s_n) = \int \ell(y_n, s_n) dT_n(x_n, dy_n)$ (conditional loss function).
- **Risk** $(T_n, \Psi_n; K_n; \theta_n) = \iint \ell(X_n(\omega), \Psi_n(K_n)) dT_n(X_n(\omega), dy_n) dK_n(\theta_n, d\omega)$ (conditional (on θ_n) average loss).

The purpose of generality

The purpose of generality here is not to solve immediate practical problems, but rather to capture the logical essence of an important concept (sufficient statistic), and in particular to disentangle that concept from such ideas as Euclidean space, dimensionality, partial differentiation, and the distinction between continuous and discrete distributions, which seem to us extraneous.

— Halmos & Savage, *Annals of Math. Stat.*, 19(2), 1948.

- Shorter proofs.
- The logical core of the argument becomes visible.
- Leverage $\sim 2.1\text{M}$ lines of formal math.

Generality of Mathlib — What is filter

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-- 'tendsto' is the generic "limit of a function" predicate.  
'tendsto f l1 l2' asserts that for every 'l2' neighborhood 'a',  
the 'f'-preimage of 'a' is an 'l1' neighborhood. -/  
def tendsto (f :  $\alpha \rightarrow \beta$ ) (l1 : filter  $\alpha$ ) (l2 : filter  $\beta$ ) := l1.map f ≤ l2
```

What is happening here is that when you are *building maths from nothing*, it's important to *isolate key concepts* and set them up in the *correct generality*. Firstly I was told what it meant for a sequence of *real or complex numbers* to tend to a limit. Then I was told what it meant for a sequence of elements of a *metric space* to tend to a limit. And finally I was told what it meant for a sequence of elements of a *topological space* to tend to a limit.

— Kevin Buzzard, *The Xena Project*, 2018 [7].

References

- [1] X. Yang *et al.*, *A Survey on Large Language Models for Mathematics: Tasks, Methods, and Challenges*. arXiv:2601.13209, 2026.
- [2] T. Tao, *Machine-Assisted Proof*. Notices Amer. Math. Soc. **72**(1), 6–13, January 2025. Based on AMS Colloquium Lecture, JMM 2024.
- [3] B. Alexeev, ChatGPT, Lean, and D. G. Mixon, *Forbidden Sidon subsets of perfect difference sets, featuring a human-assisted proof*. arXiv preprint, 2026.
- [4] R. Degenne *et al.*, *Markov kernels in Mathlib*. arXiv:2510.04070, 2025.
- [5] P. R. Halmos and L. J. Savage, *Application of the Radon–Nikodym Theorem to the Theory of Sufficient Statistics*. Ann. Math. Statist. **19**(2), 225–241, 1948.
- [6] L. Le Cam, *Asymptotic Methods in Statistical Decision Theory*. Springer-Verlag, New York, 1986.
- [7] K. Buzzard, *What is a filter? — How some computer scientists think about limits*. The Xena Project, 4 August 2018. <https://xenaproject.wordpress.com/2018/08/04/what-is-a-filter-how-some-computer-scientists-think-about-limits/>

Online resources

- Mathlib documentation: https://leanprover-community.github.io/mathlib4_docs
- Mathlib live statistics: https://leanprover-community.github.io/mathlib_stats.html
- Mathlib overview: <https://leanprover-community.github.io/mathlib-overview.html>
- Testing-lower-bounds blueprint (R. Degenne): <https://remydegenne.github.io/testing-lower-bounds/blueprint/>
- Sphere-packing in Lean (Math, Inc. Gauss agent): <https://www.math.inc/sphere-packing>

Collaborators



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Thank you.

The risk: AI slop in the library

Autoformalization is a tool, not a strategy. Autoformalizing statistics in a vacuum is actively harmful.

- LLMs can produce thousands of plausible Lean definitions per day.
- Most will type-check. Few will be the *right* definition in context — the one that composes with what's already in mathlib.
- The bottleneck of the formalization is review, not production. Flooding the queue with auto-generated PRs makes the queue worse, not better.
- *Kim Morrison*: "...without explicit plans for getting past 'merely true' all the way up to 'useful to others.' "